KING SALMAN INTERNATIONAL UNIVERSITY

Faculty of Engineering

Lectures : (2)

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Dr : Alaa Adam

Vectors

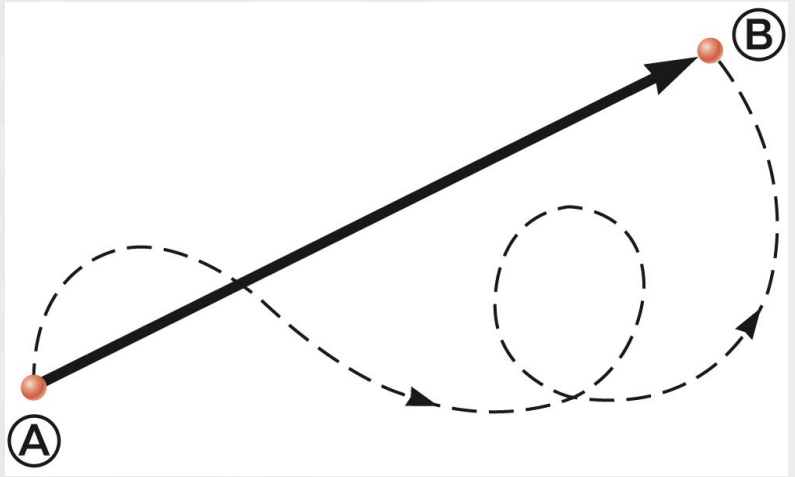
- Vector quantities
  - Physical quantities that have both numerical and directional properties
- Mathematical operations of vectors in this chapter
  - Addition
  - Subtraction

Vectors and Scalars

- A ***scalar quantity*** is completely specified by a single value with an appropriate unit and has no direction.
  - Many are always positive
  - Some may be positive or negative
  - Rules for ordinary arithmetic are used to manipulate scalar quantities.
- A ***vector quantity*** is completely described by a number and appropriate units plus a direction.

Vector Example

- A particle travels from A to B along the path shown by the broken line.
  - This is the ***distance*** traveled and is a scalar.
- The ***displacement*** is the solid line from A to B
  - The displacement is independent of the path taken between the two points.
  - Displacement is a vector.

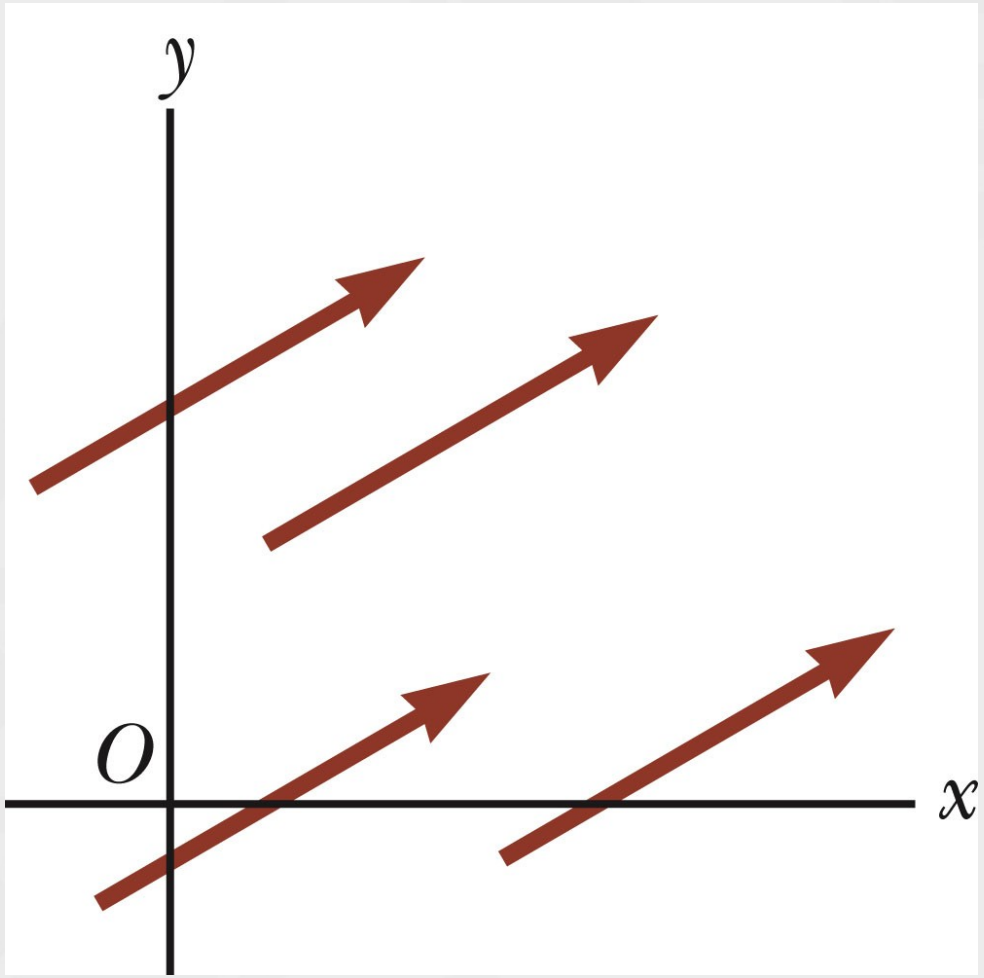


Vector Notation

- Text uses bold with arrow to denote a vector: $\vec{\mathbf{A}}$
- Also used for printing is simple bold print: **A**
- When dealing with just the magnitude of a vector in print, an italic letter will be used: A or $|\mathbf{A}|$
  - The magnitude of the vector has physical units.
  - The magnitude of a vector is always a positive number.
- When handwritten, use an arrow: $\vec{A}$

Equality of Two Vectors

- Two vectors are ***equal*** if they have the same magnitude and the same direction. $\mathbf{A} = \mathbf{B}$
- if $A = B$ and they point along parallel lines
- All of the vectors shown are equal.
- Allows a vector to be moved to a position parallel to itself



Adding Vectors

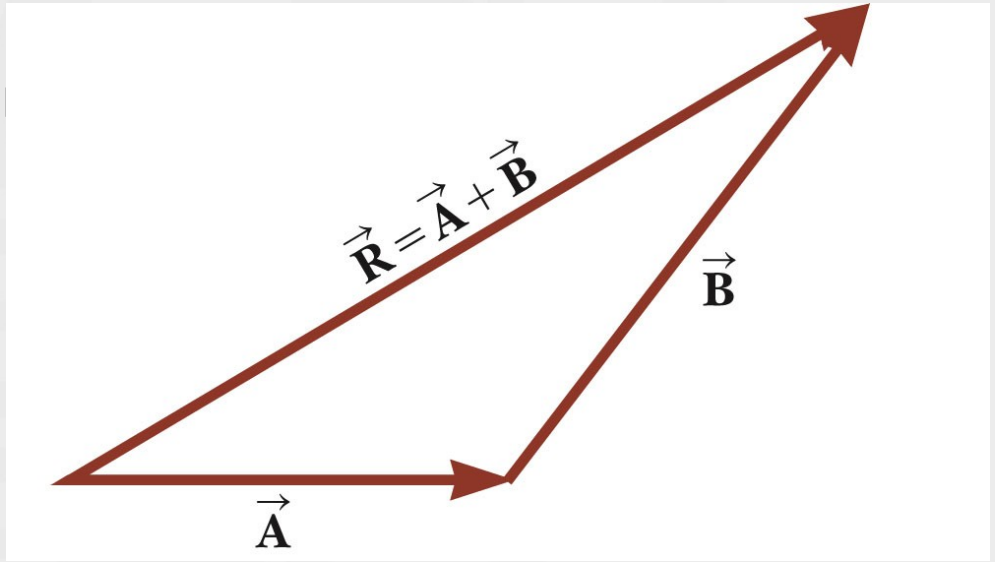
- Vector addition is very different from adding scalar quantities.
- When adding vectors, their directions must be taken into account.
- Units must be the same
- Graphical Methods
  - Use scale drawings
- Algebraic Methods
  - More convenient

Adding Vectors Graphically

- Choose a scale.
- Draw the first vector, $\mathbf{A}$, with the appropriate length and in the direction specified, with respect to a coordinate system.
- Draw the next vector with the appropriate length and in the direction specified, with respect to a coordinate system whose origin is the end of vector $\mathbf{A}$ and parallel to the coordinate system used for $\mathbf{A}$.

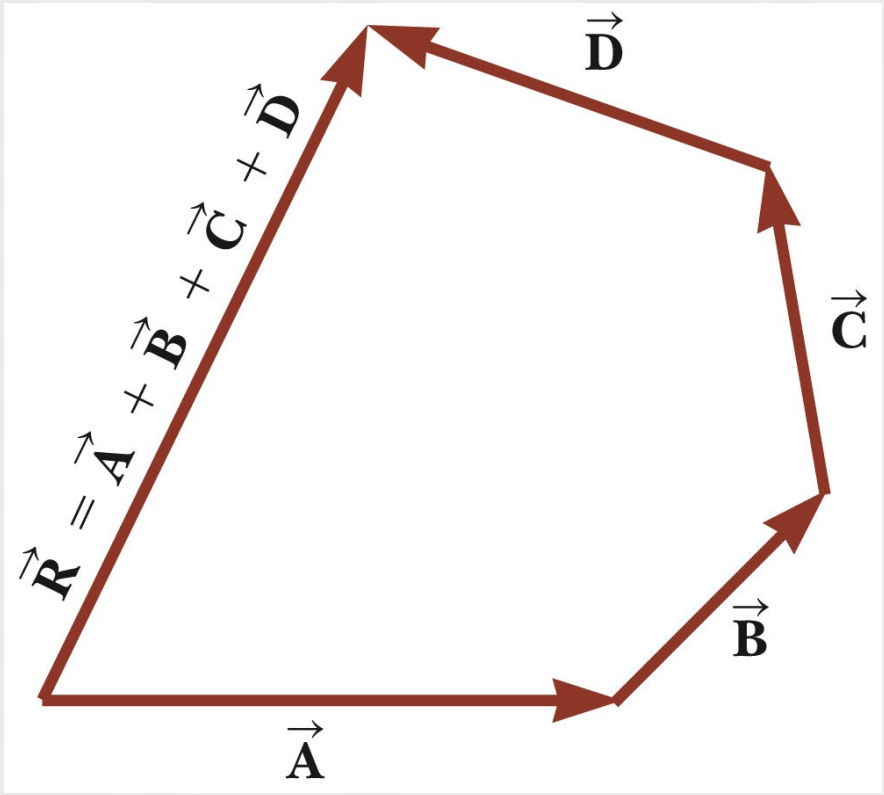
Adding Vectors Graphically, cont.

- Continue drawing the vectors “tip-to-tail” or “head-to-tail”.
- The resultant is drawn from the origin of the first vector to the end of the last vector.
- Measure the length of the resultant and its angle.
  - Use the scale factor to conve length to actual magnitude.



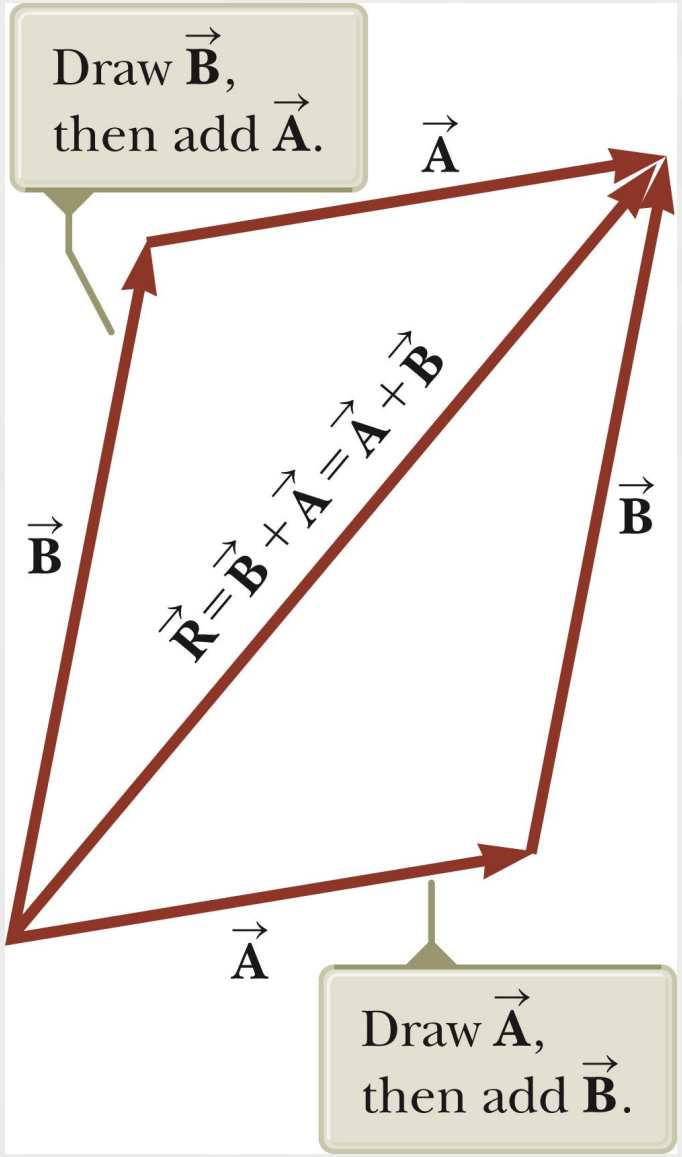
Adding Vectors Graphically, final

- When you have many vectors, just keep repeating the process until all are included.
- The resultant is still drawn from the tail of the first vector to the tip of the last vector.



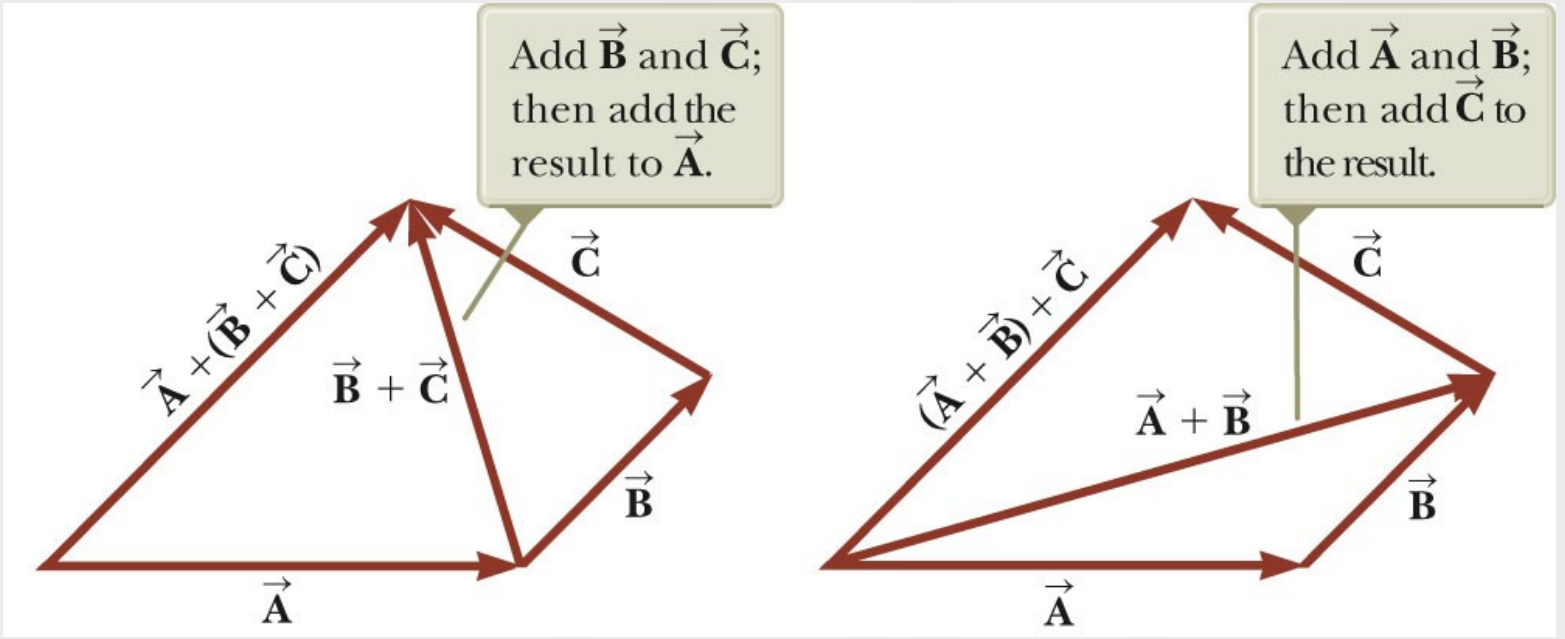
Adding Vectors, Rules

- When two vectors are added, the sum is independent of the order of the addition.
  - This is the ***Commutative Law of Addition.***
- $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$



Adding Vectors, Rules cont.

- When adding three or more vectors, their sum is independent of the way in which the individual vectors are grouped.
  - This is called the ***Associative Property of Addition.***
- $\mathbf{A}+(\mathbf{B}+\mathbf{C})=(\mathbf{A}+\mathbf{B})+\mathbf{C}$



Adding Vectors, Rules final

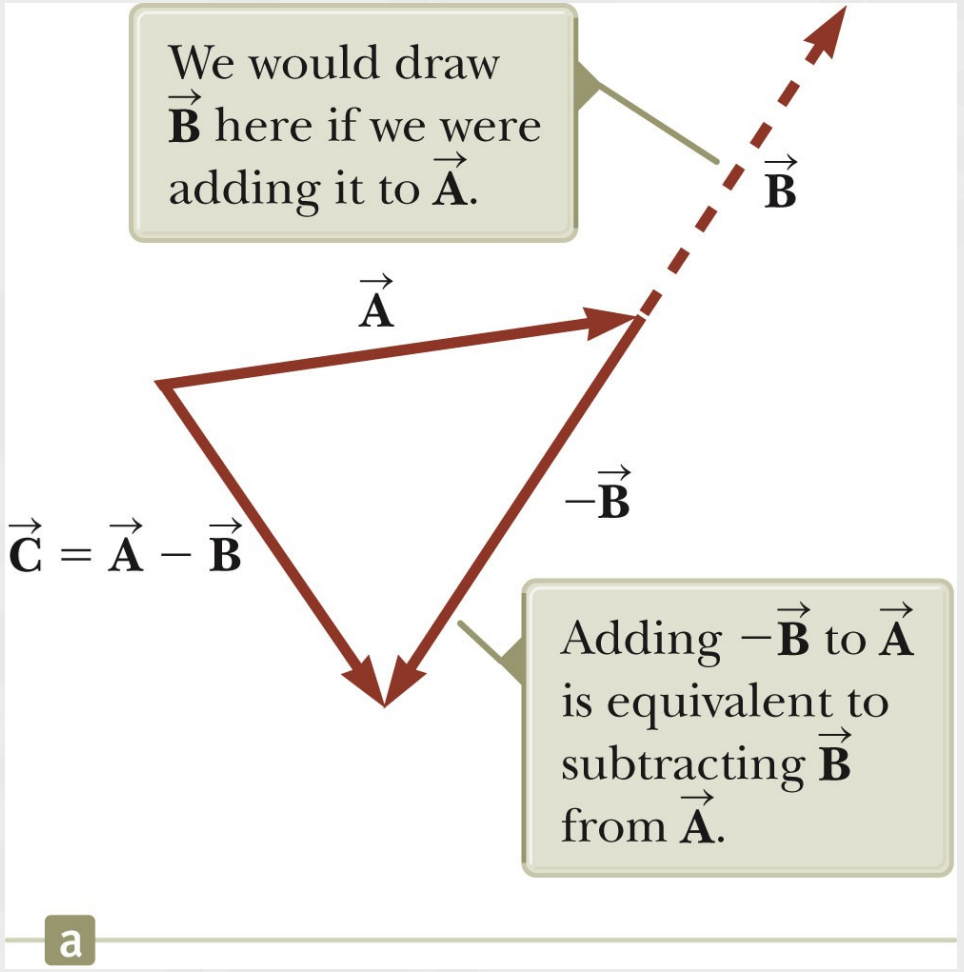
- When adding vectors, all of the vectors **must have the same units**.
- All of the vectors must be of the **same type of quantity**.
  - For example, you cannot add a displacement to a velocity.

Negative of a Vector

- The negative of a vector is defined as the vector that, when added to the original vector, gives a resultant of zero.
  - Represented as $\mathbf{A} + (-\mathbf{A}) = 0$
  -
- The negative of the vector will have the same magnitude, but point in the opposite direction.

Subtracting Vectors

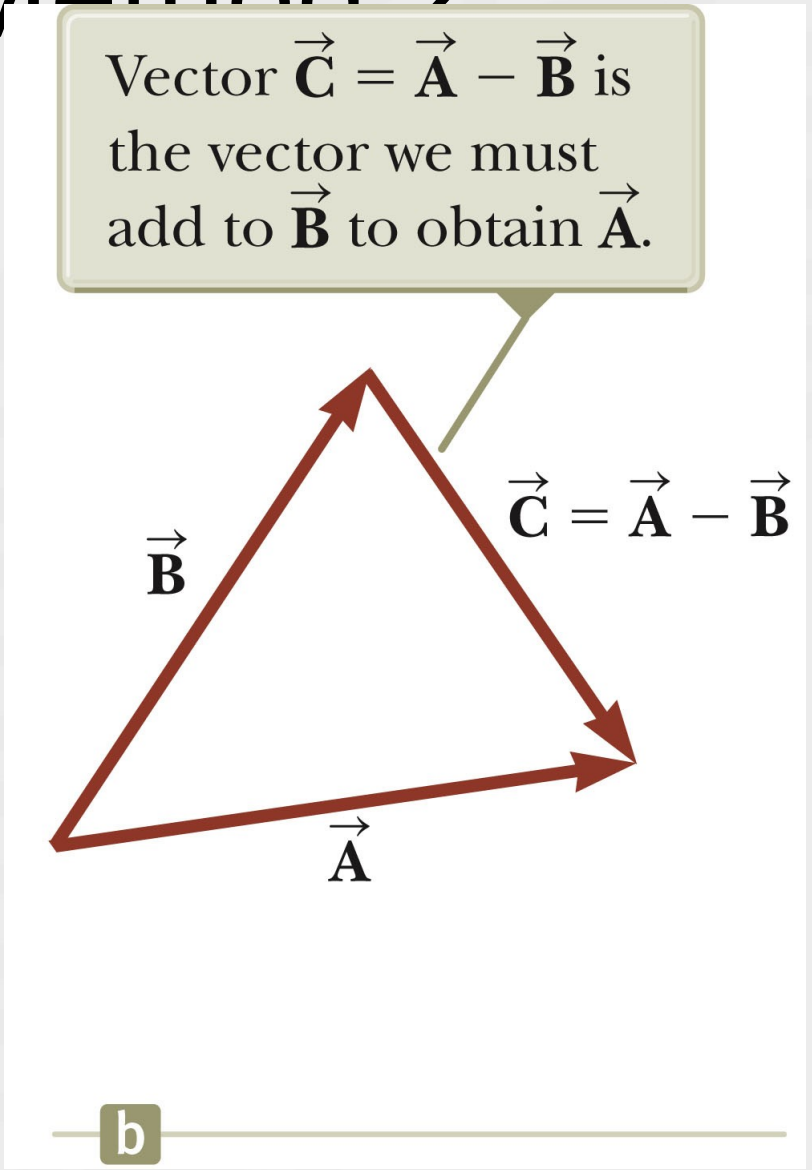
- Special case of vector addition:
- If $A - B$, then use $A + (-B)$
- Continue with standard vector addition procedure.



Subtracting Vectors, Method 2

•Another way to look at subtraction is to find the vector that, added to the second vector gives you the first vector.

- $\mathbf{A} + (-\mathbf{B}) = \mathbf{C}$
  - As shown, the resultant vector points from the tip of the second to the tip of the first.



Multiplying or Dividing a Vector by a Scalar

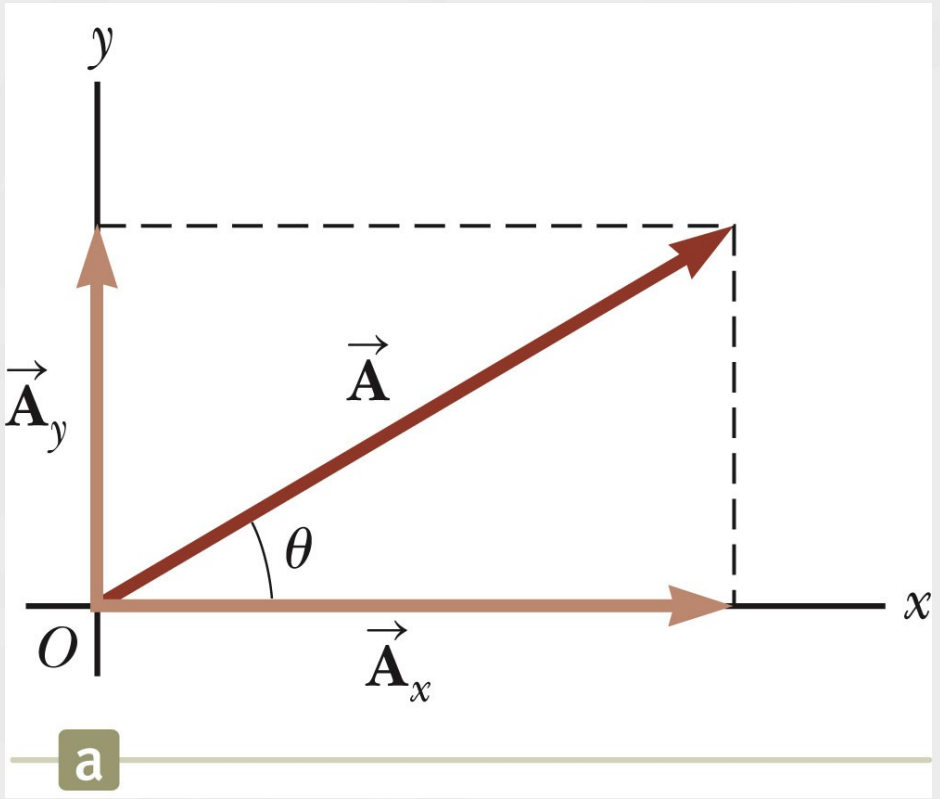
- The result of the multiplication or division of a vector by a scalar is a **vector**.
- The magnitude of the vector is multiplied or divided by the scalar.
- If the scalar is positive, the direction of the result is the same as of the original vector.
- If the scalar is negative, the direction of the result is opposite that of the original vector.

Component Method of Adding Vectors

- Graphical addition is not recommended when:
  - High accuracy is required
  - If you have a three-dimensional problem
- **Component method** is an **alternative** method
  - It uses projections of vectors along coordinate axes

Components of a Vector, Introduction

- A **component** is a projection of a vector along an axis.
  - Any vector can be completely described by its components.
- It is useful to use **rectangular components.**
  - These are the projections of the vector along the x- and y-axes.



Vector Component Terminology

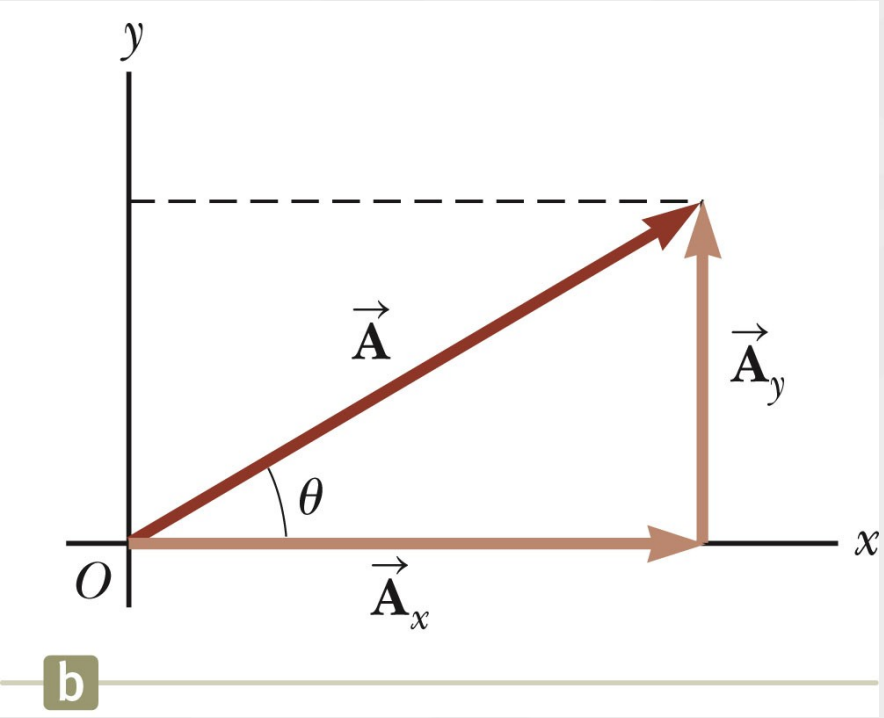
- $\mathbf{A}_x$ and $\mathbf{A}_y$ are the ***component vectors*** of $\mathbf{A}$.
  - They are vectors and follow all the rules for vectors.
- A_x and A_y are scalars, and will be referred to as the ***components*** of $\mathbf{A}$.

Components of a Vector

- Assume you are given a vector $\mathbf{A}$
- It can be expressed in terms of two other vectors, $\mathbf{A}_x$ and $\mathbf{A}_y$

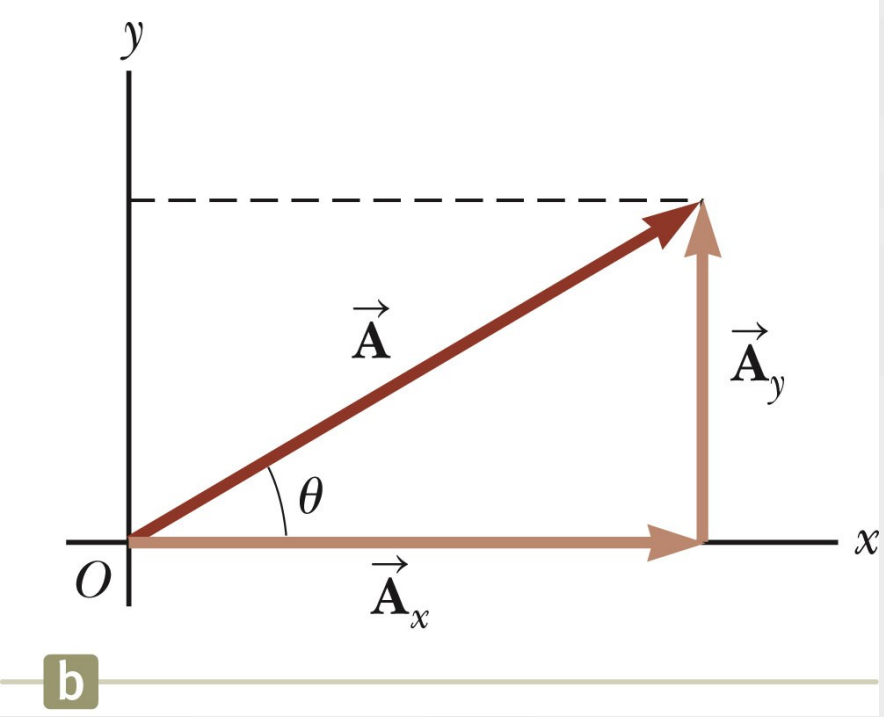
$$\mathbf{A} = \mathbf{A}_x + \mathbf{A}_y$$

- These three vectors form a right triangle.
-



Components of a Vector, 2

- The *y*-component is moved to the end of the *x*-component.
- This is due to the fact that any vector can be moved parallel to itself without being affected.
  - This completes the triangle.



Components of a Vector, 3

- The x-component of a vector is the projection along the x-axis.

$$A_x = A\cos\theta$$

- The y-component of a vector is the projection along the y-axis.

$$A_y = A\sin\theta$$

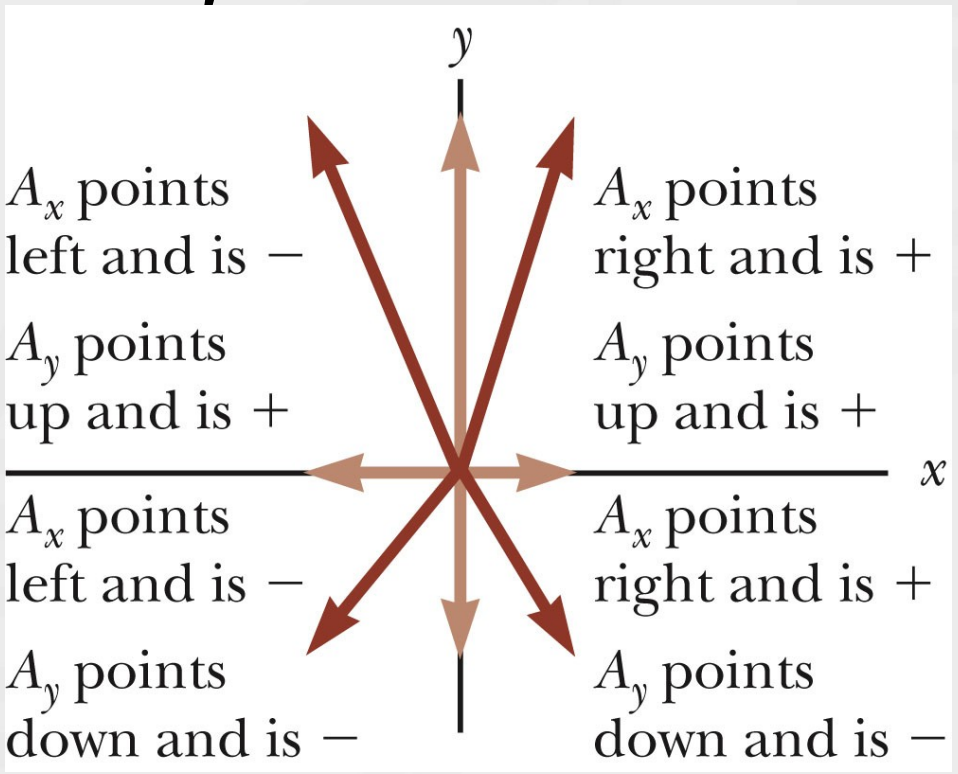
- This assumes the angle θ is measured with respect to the x-axis.
  - If not, do not use these equations, use the sides of the triangle directly.

Components of a Vector, 4

- The components are the legs of the right triangle whose hypotenuse is the length of *A*.
  - $A = \sqrt{A_x^2 + A_y^2}$ and $\theta = \tan^{-1}\frac{A_y}{A_x}$
  - May still have to find θ with respect to the positive *x*-axis
- In a problem, a vector may be specified by its components or its magnitude and direction.

Components of a Vector, final

- The components can be positive or negative and will have the same units as the original vector.
- The signs of the components will depend on the angle.

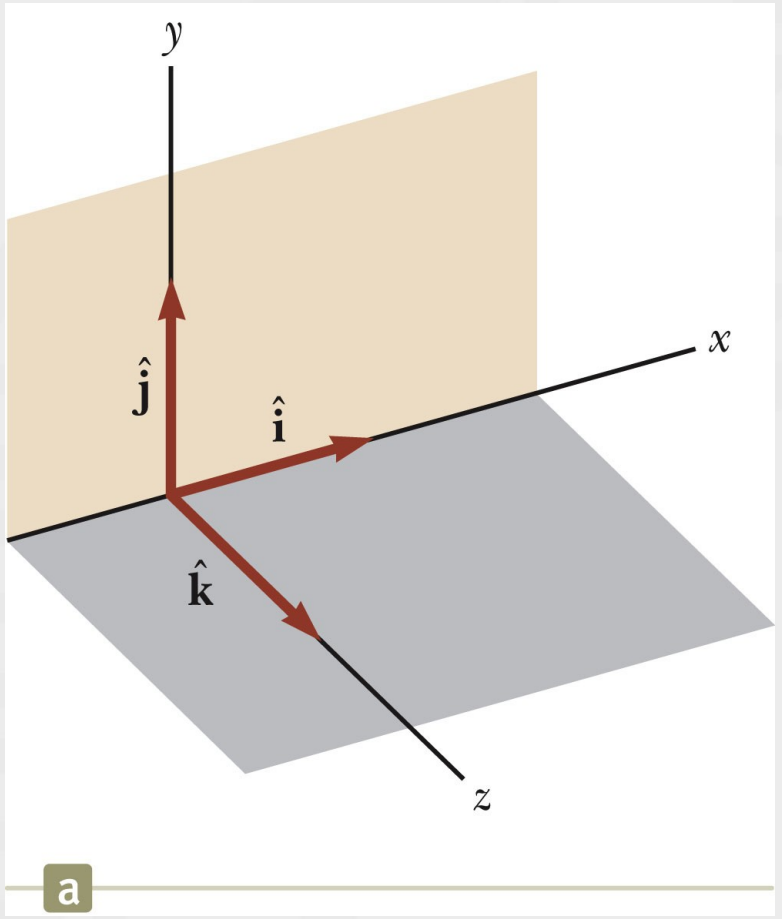


Unit Vectors

- A ***unit vector*** is a dimensionless vector with a magnitude of exactly 1.
- Unit vectors are used to specify a direction and have no other physical significance.

Unit Vectors, cont.

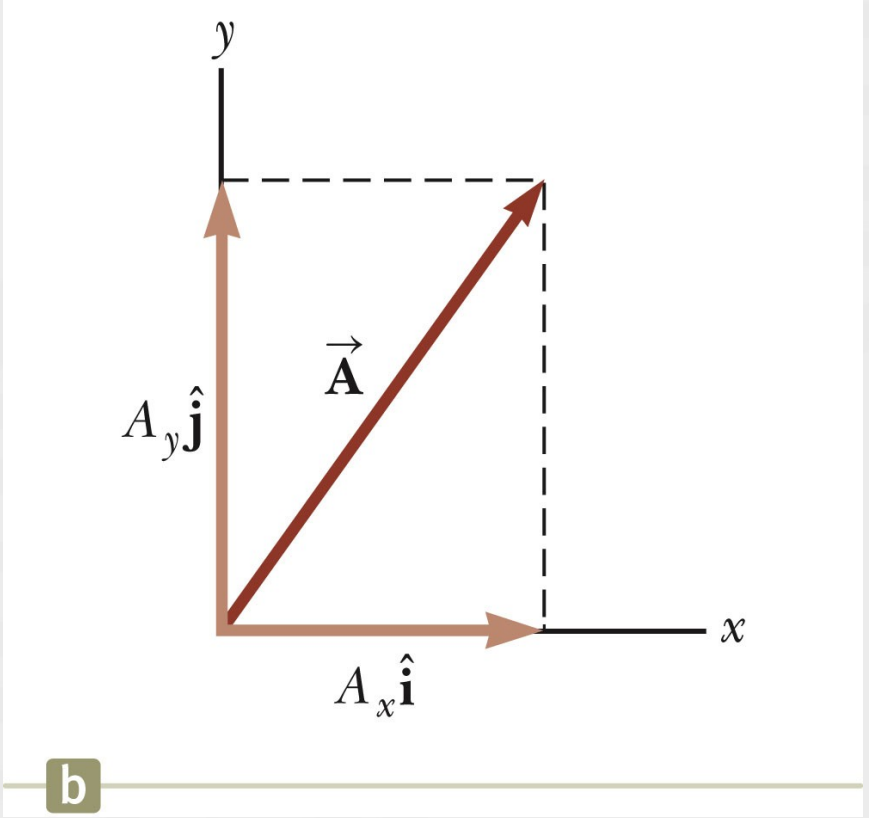
- The symbols $\hat{i}, \hat{j}$, and $\hat{k}$ represent unit vectors
- They form a set of mutually perpendicular vectors in a right-handed coordinate system $|\hat{i}| = |\hat{j}| = |\hat{k}| = 1$
- The magnitude of each unit vector is 1



Unit Vectors in Vector Notation

- $\mathbf{A}_x$ is the same as $A_x\,\hat{\mathbf{i}}$ and $\mathbf{A}_y$ is the same as $A_y\,\hat{\mathbf{j}}$ etc.
- The complete vector can be expressed as:

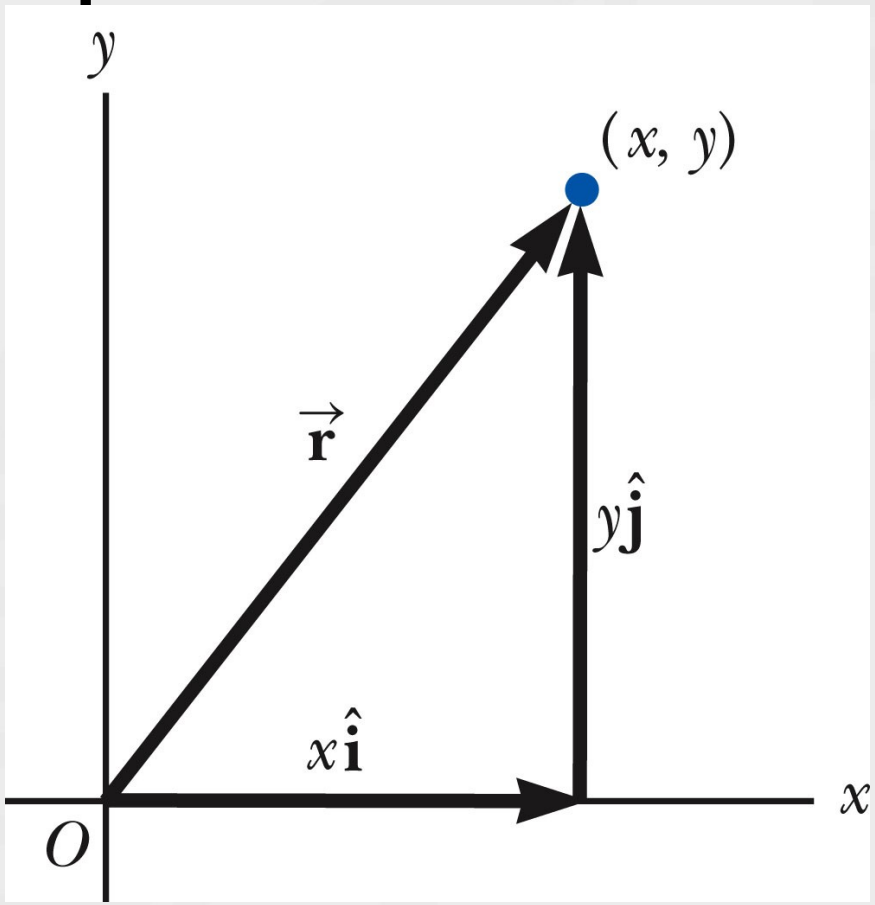
$$\mathbf{A} = A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}}$$



Position Vector, Example

- A point lies in the *xy* plane and has Cartesian coordinates of (x, y).
- The point can be specified by the position vector.
- This gives the components of the vector and its coordinates.

$$\hat{\mathbf{r}} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$$



Adding Vectors Using Unit Vectors

- Using $\mathbf{R} = \mathbf{A} + \mathbf{B}$
- Then

$$\vec{\mathbf{R}} = (A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}}) + (B_x \hat{\mathbf{i}} + B_y \hat{\mathbf{j}})$$

$$\vec{\mathbf{R}} = (A_x + B_x)\hat{\mathbf{i}} + (A_y + B_y)\hat{\mathbf{j}}$$

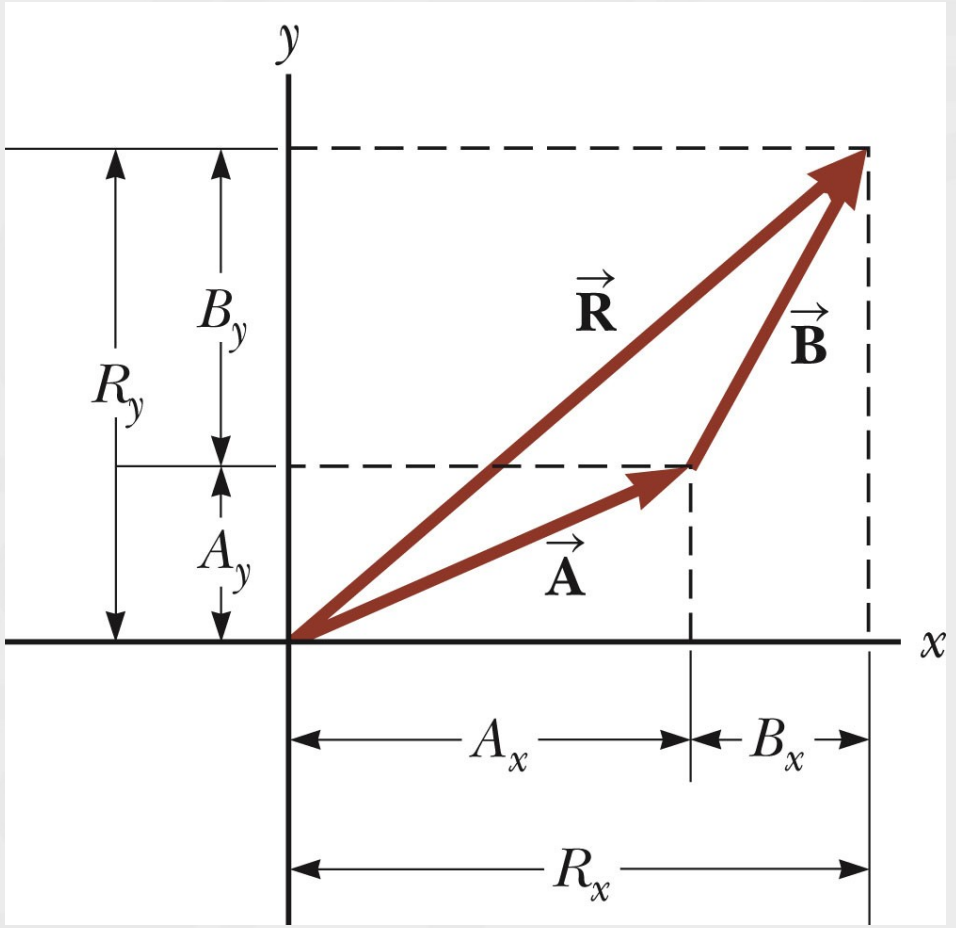
$$\mathbf{R} = R_x \hat{\mathbf{i}} + R_y \hat{\mathbf{j}}$$

- So $R_x = A_x + B_x$ and $R_y = A_y + B_y$

$$R = \sqrt{R_x^2 + R_y^2} \qquad \theta = \tan^{-1} \frac{R_y}{R_x}$$

Adding Vectors with Unit Vectors

- Note the relationships among the components of the resultant and the components of the original vectors.
- $R_x = A_x + B_x$
- $R_y = A_y + B_y$



Three-Dimensional Extension

- Using $\mathbf{R} = \mathbf{A} + \mathbf{B}$
- Then

$$\vec{\mathbf{R}} = (A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}} + A_z\hat{\mathbf{k}}) + (B_x\hat{\mathbf{i}} + B_y\hat{\mathbf{j}} + B_z\hat{\mathbf{k}})$$

$$\vec{\mathbf{R}} = (A_x + B_x)\hat{\mathbf{i}} + (A_y + B_y)\hat{\mathbf{j}} + (A_z + B_z)\hat{\mathbf{k}}$$

$$\mathbf{R} = R_x\hat{\mathbf{i}} + R_y\hat{\mathbf{j}} + R_z\hat{\mathbf{k}}$$

- So $R_x = A_x + B_x$, $R_y = A_y + B_y$, and $R_z = A_z + B_z$

$$R = \sqrt{R_x^2 + R_y^2 + R_z^2} \qquad \theta_x = \cos^{-1}\frac{R_x}{R}, \textit{etc.}$$

Adding Three or More Vectors

- The same method can be extended to adding three or more vectors.
- Assume $\mathbf{R} = \mathbf{A} + \mathbf{B} + \mathbf{C}$

$$\mathbf{R} = (A_x + B_x + C_x)\hat{\mathbf{i}} + (A_y + B_y + C_y)\hat{\mathbf{j}}$$

- And $+ (A_z + B_z + C_z)\hat{\mathbf{k}}$

Compute the vector's magnitude: $\vec{A}$ = 5i – 4j + 2k.

$$\sqrt{45}$$

Two vectors $\vec{A}$ = 3i + 2j and $\vec{B}$ = -i + 4j

Find A + B.



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THANK YOU

16949 www.ksiu.edu.eg info@ksiu.edu.eg

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